

Abdul Raffay

Senior Python & Django Engineer | AI/LLM Systems | Full-Stack Backend
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PROFESSIONAL SUMMARY

Senior Python/Django Engineer with 5+ years building scalable SaaS platforms, real-time systems, and enterprise-grade web applications across broadcast media, government, and startup environments. Deep expertise in Django REST Framework, high-concurrency live voting systems, microservices, and LLM integration (LangChain, RAG, OpenAI). Proven track record delivering mission-critical systems at scale — from live TV platforms handling 100K+ concurrent viewers (Banijay Nordic/Tellybug) to military hospital networks across Pakistan.

WORK EXPERIENCE

Senior Software Engineer — Python & Django | [Adappt Ltd](#)

Nov 2025 – Present

- Worked on Tellybug, a Django-based interactive TV platform powering Banijay Nordic Mobile second-screen apps for live broadcast shows (X Factor, Allvone, GameJamm) handling 10K-100K concurrent viewers per episode.
- Built and maintained live audience voting systems — vote APIs, poll state machines, result freezing/reveal flows, leaderboards, and high-concurrency capacity handling — backed by Redis, DynamoDB, Celery, and cached state versioning.
- Operated Allvone ('All Against One') show runtime: producer console driving real-time round flow (numeric → numeric_ending → numeric_closed → numeric_reveal → round_end), numeric vote ingestion, playlist-driven timelines, and state sync to mobile clients.
- Extended X Factor interactive features — multichoice, rater, decision-time, trivia, and predictor poll states, vote endpoints, reveal flows, playalong alerts, and console integrations for production teams during live episodes.
- Developed GameJamm, a multi-tenant SaaS layer — org workspaces, subscriptions/RBAC, video templates, and game sessions that publish into the Allvone voting runtime for customizable interactive games.
- Contributed to LLM evaluation pipelines — designed test harnesses and custom metrics to assess model behavior across safety, factuality, and instruction-following dimensions for open-source and proprietary models.

Python Django Developer | [Ministry of Defence, Govt. of Pakistan](#)

Feb 2022 – Nov 2025

- Led end-to-end development of an e-Health Management System deployed across 50+ military hospitals in Pakistan, digitizing patient records, billing, appointments, and financial reporting for 10,000+ users.
- Designed and built a Medical Tourism SaaS application using Django, Vue.js, and PostgreSQL — integrating a secure payment gateway, multi-vendor booking (transport, hotels, visas), and a real-time sales dashboard.
- Developed a full-stack GitHub analytics platform using Flask, React, PostgreSQL, and Docker for organizational code quality tracking.
- Delivered a Network Automation Tool using Django REST Framework and React.js for real-time monitoring and management of data center networking devices, including live device performance metrics and traffic analytics.
- Enforced change control best practices across all deployments, ensuring zero-downtime releases for production systems.

Python Django Developer | [Epochs IT Solutions](#)

Aug 2020 – Feb 2022

- Managed and improved existing client applications and delivered new full-stack web apps using Django, React, and PostgreSQL.

- Designed database schemas, built RESTful APIs, and implemented frontend features in cross-functional team settings.
- Identified and implemented system enhancements to optimize performance and streamline administrative workflows.

KEY PROJECTS

Dealmart | *Python, Django, DRF, React, Celery, Redis, MySQL, OpenAI, scikit-learn, Plaid, Twilio*

- Built a multi-tenant SaaS platform for 50-200 automotive dealerships powering CRM, financing, inventory management, and AI-driven sales tools.
- Integrated OpenAI GPT-4o for personalized SMS outreach, an AI reply agent with CRM context and inventory search, and on-site banner personalization.
- Implemented in-house ML models (scikit-learn) for finance approval prediction and stock risk scoring using FICO scores, income, and credit features.
- Integrated Plaid for identity verification, CRA credit reports, and cashflow analysis; Twilio/SendGrid for SMS/email; S3-compatible storage for documents.

DocPilot | *Python, Django, Vue.js, PostgreSQL, Docker, LangChain, Ollama, HuggingFace*

- Built a full-stack AI document chat system allowing users to upload PDFs, CSVs, and XLS files and receive context-aware answers powered entirely by local LLMs — ensuring complete data privacy with zero cloud inference.
- Implemented RAG pipeline using LangChain, HuggingFace embeddings, and PostgreSQL vector storage (pgvector) for fast, accurate document retrieval.

e-Health Management System | *Python, Django, Vue.js, PostgreSQL, Docker*

- Delivered an enterprise health platform digitizing patient registration, medical records, appointments, billing, and audit reporting — now actively running across 50+ military hospitals in Pakistan, serving thousands of clinical and administrative staff.
- Ensured scalability, data security, and role-based access control for sensitive medical data across multiple hospital departments.

Tellybug — Interactive TV Platform | *Python, Django, DRF, Celery, MySQL, Redis, DynamoDB, RabbitMQ, AWS, Memcached*

- Cloud backend for Banijay Nordic Mobile second-screen TV apps — powering live audience voting, producer operator consoles, and show state orchestration for formats including Allvone, X Factor, and GameJamm-powered games at 100K+ concurrent viewers per live broadcast.
- Show formats covered: Allvone (numeric guessing with real-time state machine), X Factor (rater, multichoice, trivia, predictor, and reveal flows), and GameJamm (multi-tenant SaaS for publishing custom video-driven game sessions into the live voting runtime).

CRM Connect | *Python, Django, React, PostgreSQL, Docker*

- Built a multi-tenant CRM featuring project and lead management, permission-based user roles, and recycle bin functionality using React.js and Django REST Framework.

The Social Wing | *Python, Django, React, PostgreSQL, Docker*

- Developed a social media management tool enabling users to schedule posts across Facebook, Instagram, Twitter, and LinkedIn, with a calendar view for planning and engagement analytics dashboard.

Friday Chatbot | *Python, Django, React, PostgreSQL, Docker*

- Built a chatbot creation service allowing users to deploy AI-powered bots for Facebook Pages, Slack channels, and Telegram accounts via React.js and Django REST Framework.

Student CRM | *Python, Django, React, PostgreSQL, Docker*

- Contributed to a consultancy firm portal with lead management and student tracking features using Django and Bootstrap-based frontend.

Resource Management System | *Python, Django, Vue.js, PostgreSQL, Docker*

- Built an end-to-end asset and equipment management system with automated checklist compliance, centralized records, and real-time dashboards for performance tracking and operational insights.

Network Automation Tool | *Python, Django, React, PostgreSQL, Docker*

- Developed a data center monitoring application with real-time device status tracking, performance metrics, and network traffic analytics using React.js frontend and Django REST Framework backend.

TECHNICAL SKILLS

Backend: Python, Django, Django REST Framework, FastAPI, Flask

AI / LLM: LangChain, RAG, HuggingFace, Ollama, OpenAI (GPT-4o), Vector Embeddings, Local LLMs

Databases: PostgreSQL, MySQL, MongoDB, Redis, Memcached

Frontend: React.js, Vue.js, HTML/CSS, JavaScript

DevOps & Cloud: AWS, Docker, Nginx, Gunicorn, Celery, Git, CI/CD

Data & Automation: pandas, NumPy, Matplotlib, scikit-learn, Scrapy, BeautifulSoup, Selenium, Netmiko, Paramiko

Languages: Python, JavaScript, SQL, C/C++, HTML/CSS

EDUCATION

Bachelor of Science in Computer Science

Bahria University, Islamabad Campus

Feb 2016 – Feb 2020